

Report HW 4 ECE-111

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Homework 4a (LFSR)

1. SystemVerilog design code

a. lfsr (student added part)

```
//student to add implementation for LFSR code
// Internal shift register
logic [N-1:0] lfsr_reg, lfsr_next, taps;
logic [$clog2((1 << N)) - 1:0] count;
logic xor_out, cycle_done;

always_comb begin
    case (N)
        2: taps = 2'b11;
        3: taps = 3'b110;
        4: taps = 4'b1100;
        5: taps = 5'b10100;
        6: taps = 6'b110000;
        7: taps = 7'b1100000;
        8: taps = 8'b10111000;
        default: taps = '0;
    endcase
end

always_comb begin
    xor_out = ^(lfsr_reg & taps);
    lfsr_next = {lfsr_reg[N-2:0], xor_out};
end

always_ff @(posedge clk or negedge reset) begin
    if (!reset) begin
        lfsr_reg <= 0;
        count <= 0;
    end else if (load_seed) begin
        lfsr_reg <= seed_data;
        count <= 1;
    end else begin
        lfsr_reg <= lfsr_next;
        if (count != 0)
            count <= count + 1;
    end
end

assign cycle_done = (count == (2**N - 1));
assign lfsr_done = cycle_done;

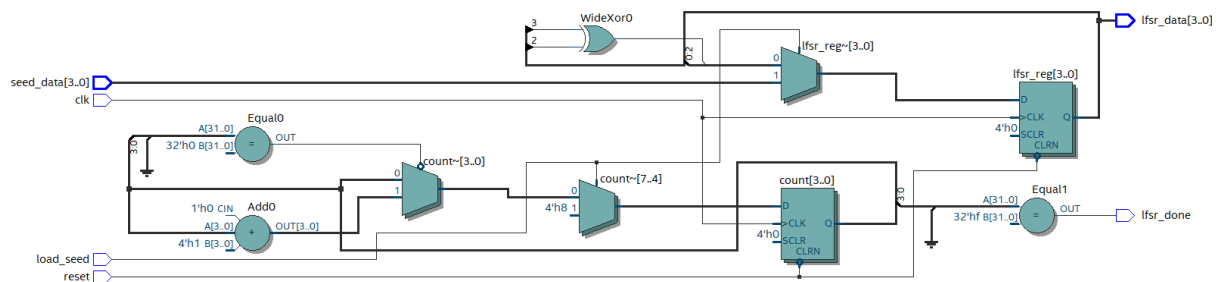
assign lfsr_data = lfsr_reg;
```

2. Synthesis Resource usage and schematic generated from RTL netlist viewer

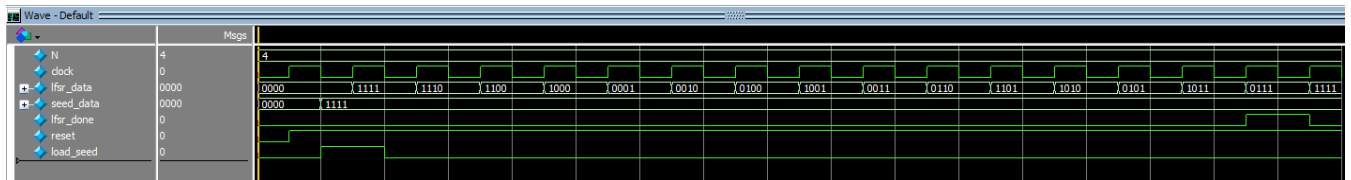
a. Resource Usage

Analysis & Synthesis Resource Usage Summary		
<<Filter>>		
	Resource	Usage
1	Estimate of Logic utilization (ALMs needed)	5
2		
3	▼ Combinational ALUT usage for logic	9
1	-- 7 input functions	0
2	-- 6 input functions	0
3	-- 5 input functions	2
4	-- 4 input functions	3
5	-- <=3 input functions	4
4		
5	Dedicated logic registers	8
6		
7	I/O pins	12
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	load_seed~input
12	Maximum fan-out	8
13	Total fan-out	75
14	Average fan-out	1.83

b. RTL netlist viewer

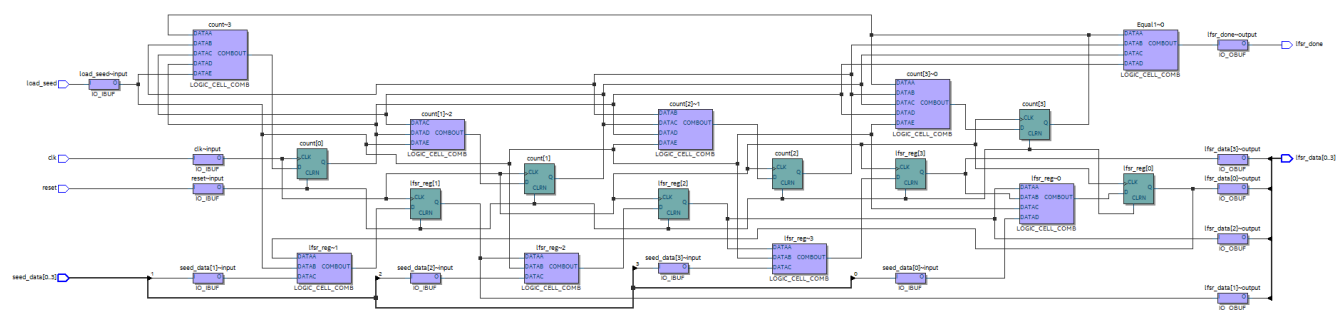


3. Simulation snapshot and explain simulation result



This simulation result shows a full 15 cycles for an $N = 4$ where the random pattern follows the feedback polynomial of $x^4 + x^3 + 1$ with max length of 15, which matches the cycle in the above case that goes through 15 different lfsr data.

4. Post-Mapping schematic



Homework 4b (Barrel Shifter)

1. SystemVerilog code snapshot

a. barrel_shifter

```
// Students to add code for barrel shifter
logic [3:0] shift0_in1, shift0;
logic [3:0] shift1_in1;

always_comb begin
    for (int i = 0; i < 4; i++) begin
        if (select) begin
            shift0_in1[i] = direction ? din[(i + 3) % 4] : din[(i + 1) % 4];
        end else begin
            if (direction)
                shift0_in1[i] = (i == 0) ? 1'b0 : din[i - 1];
            else
                shift0_in1[i] = (i == 3) ? 1'b0 : din[i + 1];
        end
    end
end

genvar i;
generate
    for (i = 0; i < 4; i++) begin : mux_shift0
        mux_2x1 mux1(.sel(shift_value[0]), .in0(din[i]), .in1(shift0_in1[i]), .out(shift0[i]));
    end
endgenerate

always_comb begin
    for (int i = 0; i < 4; i++) begin
        if (select) begin
            shift1_in1[i] = direction ? shift0[(i + 2) % 4] : shift0[(i + 2) % 4];
        end else begin
            if (direction)
                shift1_in1[i] = (i <= 1) ? 1'b0 : shift0[i - 2];
            else
                shift1_in1[i] = (i >= 2) ? 1'b0 : shift0[i + 2];
        end
    end
end

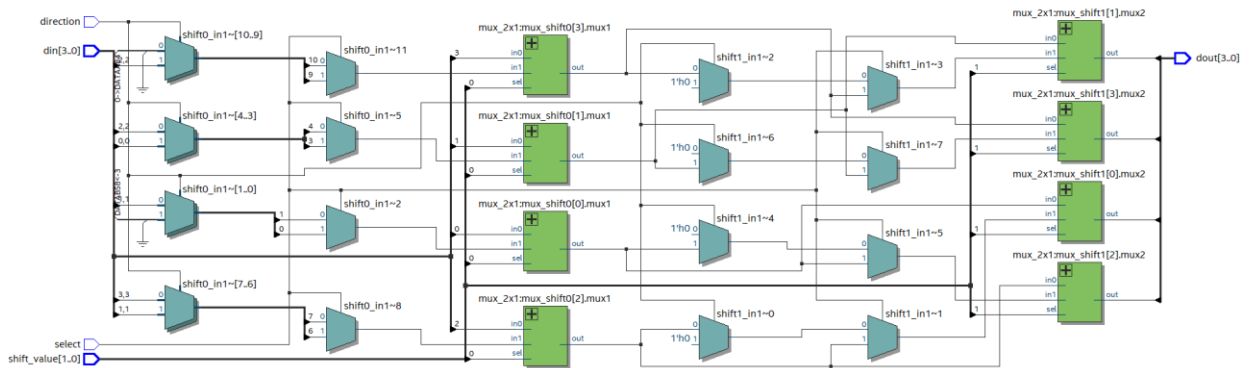
generate
    for (i = 0; i < 4; i++) begin : mux_shift1
        mux_2x1 mux2(.sel(shift_value[1]), .in0(shift0[i]), .in1(shift1_in1[i]), .out(dout[i]));
    end
endgenerate
```

2. Synthesis Resource usage and schematic generated from RTL netlist viewer

a. Synthesis Resource usage

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2		
3	▼ Combinational ALUT usage for logic	8
1	-- 7 input functions	0
2	-- 6 input functions	2
3	-- 5 input functions	6
4	-- 4 input functions	0
5	-- <=3 input functions	0
4		
5	Dedicated logic registers	0
6		
7	I/O pins	12
8		
9	Total DSP Blocks	0
10		
11	Maximum fan-out node	direction~input
12	Maximum fan-out	8
13	Total fan-out	58
14	Average fan-out	1.81

b. RTL netlist viewer



3. Simulation snapshot and explain simulation result

Wave - Default		Mags															
select	St1																
direction	St1																
shift_value	11	00	01	10	11	00	01	10	11	00	01	10	11	00	01	10	11
dn	1011	0000	1000			0000	1011			0000	0001			0000	1011		
dout	1101	0000	0100	0010	0001	0000	1101	1110	0111	0000	0010	0100	1000	0000	0111	1110	1101

The simulation wave graph shows that under select 0 or 1, for direction 0 or 1, each shift value (00,01,10,11) each din correctly output the dout with the select 0 performs shift, 1 performs rotate, while the direction 0 is to right and 1 is to left, and shift_value is how many bit positions to move. The simulated waveform shows exactly the expected output for each input combinations.

4. Post-Mapping schematic

